

SIG QUANT INTERVIEW - PREP SHEET (JING XIANG XU)

For the October 8 interview (Quantitative Research + Quantitative Systematic Trading). SIG's style: fast probability and expected value, mental math under time pressure, game theory and market-making scenarios, and probing whether you can explain your reasoning cleanly. Polish your two stories (Polymarket book; Hang Seng index prediction) and practice the math below out loud, to a timer.

1. CORE PROBABILITY & EXPECTED VALUE (THE BREAD AND BUTTER)

Q1. Two players alternate flipping a fair coin; the first to flip Heads wins. What is the probability the first player wins?

Answer: First player wins on flips 1, 3, 5, ...: $P = 1/2 + 1/8 + 1/32 + \dots = (1/2)/(1 - 1/4) = 2/3$.

Fast method: first player wins with probability p where $p = 1/2 + (1/4)p$, so $p = 2/3$.

Takeaway: Recognize geometric series or set up a self-referential equation; both take under 30 seconds.

Q2. You roll a die until a 6 appears. What is the expected number of rolls, and the expected number of rolls that show an even number?

Answer: Geometric with $p = 1/6$: $E[\text{rolls}] = 6$.

Every roll is even with probability $1/2$ regardless of length; by Wald's identity $E[\text{even rolls}] = E[N] \times P(\text{even per roll}) = 6 \times 1/2 = 3$.

Takeaway: Wald's equation shortcuts many 'expected count' questions - say the word 'Wald' to show structure.

Q3. I offer you a game: pay 10, roll a die, I pay you the square of the outcome. Do you play?

Answer: $E[\text{outcome squared}] = (1+4+9+16+25+36)/6 = 91/6 = 15.17 > 10$, so yes, $EV = +5.17$ per roll.

Follow-up they will ask: 'now I pay you the square of (outcome - 3)'. Variance view: $E[(X-3)^2] = \text{Var}(X) + (E[X]-3)^2 = 35/12 + 0.25 = 3.17$. Now you should decline if the price is 10.

Takeaway: Always decompose: $E[f(X)]$ is not $f(E[X])$. Quote variance where relevant.

Q4. A jar has 50 red and 50 blue balls. Move balls between two jars to maximize $P(\text{red when drawing uniformly from a random jar})$.

Answer: Put 1 red ball alone in jar A, the remaining 49 red + 50 blue in jar B.

$P = (1/2)(1) + (1/2)(49/99) = 1/2 + 49/198 = 148/198 = 74/99 = 0.7475$.

Takeaway: Extremize at the boundary: make one jar as red as possible.

Q5. You have a biased coin with $P(H) = p$ but you don't know p . How do you generate a fair coin flip?

Answer: Von Neumann scheme: flip pairs until they differ (HT or TH); call HT = H, TH = T. Each differing pair is fair by symmetry.

Expected pairs needed: $1/(2p(1-p)) \leq 2$.

Takeaway: Classic 'reduction to symmetry' - works for any unknown bias.

2. EXPECTED VALUE WITH DECISIONS (THE SIG SIGNATURE)

Q6. You may roll a die up to three times; after each roll take the value or reroll (last roll is forced). What is the fair value of the game and your optimal strategy?

Answer: $V1 = 3.5$. With two rolls left: stop if roll > 3.5 , so $V2 = (4+5+6)/6 \times \text{avg}(4,5,6) + (3/6) \times 3.5 = (15/6)(5)/\dots$ compute directly: $V2 = (1/6)(4+5+6) + (1/2)(3.5) = 2.5 + 1.75 = 4.25$.

With three: $V3 = (1/6)(5+6) + (2/3) \times V2 \dots$ precisely: stop if roll $> V2 = 4.25$, i.e. on 5 or 6: $V3 = (1/6)(5+6) + (4/6)(4.25) = 11/6 + 17/6 = 4.667$.

Strategy: keep 5 or 6 on roll one, keep 4+ on roll two.

Takeaway: Solve backwards (dynamic programming). State the cutoff at each stage before computing.

Q7. Five ants sit on a 1-meter plank facing left or right at random, walk at 1 m/s, and reverse direction when they collide. How long until all are off?

Answer: Equivalent (relabeling) to ants passing through each other: max time = $1 \text{ m} / 1 \text{ m/s} = 1 \text{ second}$.

Takeaway: Symmetry/relabeling tricks turn collisions into non-events - look for them constantly.

Q8. You flip 100 fair coins. Those showing H pay you \$1; then you may reflip all T coins once (all T together). Fair value?

Answer: Let k = number of heads. After the single reflip of the rest, total = $100/2 + E[\text{Bin}(100 - k, 1/2) | k] \dots$

unconditional: each coin ends H with probability $1/2 + (1/2)(1/2) = 3/4$, so $EV = 75$.

Takeaway: Linearity of expectation over independent coins beats conditioning on k .

Q9. St. Petersburg game: flip until H, paid 2^n . What is the fair price? What would you actually pay?

Answer: $E = \sum (1/2^n)(2^n) = \text{infinity}$, but utility/log-wealth or bankroll constraints give realistic values: with log utility the certainty equivalent is a small finite number (per-coin toss analysis gives ~\$2-10 range depending on assumptions).

SIG wants: state the math, then say 'in practice my counterparty's credit and my bankroll cap it near a few dollars'.

Takeaway: Never just say 'infinite' - discuss practical constraints; they test judgment, not recall.

Q10. Deal 5 cards. What is the probability of exactly one pair?

Answer: $C(13,1) \times C(4,2) \times C(12,3) \times 4^3 / C(52,5) = (13 \times 6 \times 220 \times 64) / 2,598,960 = 1,098,240 / 2,598,960 = 42.26\%$.

Structure: rank for the pair, 3 distinct other ranks, suits.

Takeaway: Know $C(52,5) = 2,598,960$ by heart; count patterns methodically.

3. MENTAL MATH & SPEED (WARM UP DAILY, TIMED)

Q11. Drill list (target: 10-15 seconds each, out loud).

Answer: 17×23 ; 24×25 ; 7.5% of 240; 61 squared; $1/17$ to 4 decimals; 2^{15} ; $C(10,3)$.

Answers: 391; 600; 18; 3721; 0.0588; 32,768; 120.

Tricks: $17 \times 23 = (20-3)(20+3) = 400-9$; $x25 = x100/4$; $a\%$ of $b = b\%$ of a ; squares via $(a+b)^2$.

Takeaway: SIG interviews include rapid-fire arithmetic; accuracy at speed beats slow perfection.

Q12. Estimation (Fermi) example: How many tennis balls fit in this room?

Answer: Volume ratio with packing factor ~64%: room $4 \times 5 \times 3 \text{ m} = 60 \text{ m}^3$; ball $d = 6.7 \text{ cm}$, $V = 1.57 \times 10^{-4} \text{ m}^3$; count = $60 \times 0.64 / 1.57 \times 10^{-4} \approx 245,000$.

State assumptions before arithmetic; round aggressively; sanity-check the order of magnitude.

Takeaway: They grade your estimation process and confidence, not the last digit.

4. GAMES, MARKET-MAKING & BEHAVIORAL SCENARIOS

Q13. Market-making: 'Make me a market on the number of windows in this building.'

Answer: Give a two-way quote with width: '35 - 55'. If pressed tighter: '42 - 48'. If they trade, update your belief: they buy at 55 only if they know something - shade up to '55 - 75'.

Never quote a point value; always show the bid-ask and explain the information update after a trade.

Takeaway: This tests Bayesian updating under adverse selection. Practice saying quotes out loud with conviction.

Q14. Guess 2/3 of the average of all guesses (0-100). What do you submit?

Answer: Iterated dominance: 0 is the Nash equilibrium, but real humans play ~20-35. Against a mixed group submit ~18-22; against quant-savvy competitors submit lower (10-15).

Say: 'equilibrium is 0, but I'm playing against people, so I expect one or two levels of reasoning, not infinite.'

Takeaway: SIG loves p-beauty-contest questions; calibrating to the population IS the skill.

Q15. Poker (they know you play): you hold a flush draw on the flop, pot 100, villain bets 50. Call?

Answer: 9 outs twice: ~36% equity by the river (rule of 4). Need $50 / (100 + 50 + 50) = 25\%$ breakeven with implied odds - clear call.

Connect it to your Polymarket edge: 'pricing gaps above 3 points' is the same idea - EV per unit risk vs price.

Takeaway: Expect one poker/probability-in-games question since it's on your resume; have numbers, not vibes.

Q16. You and a rival simultaneously choose a number 1-5; higher number wins \$10 from the other unless the difference is 1, in which case the lower number wins. What do you play?

Answer: Mixed-strategy game; pure strategies lose to counters (2 beats 1 and 5 is beaten by 4...). Equilibrium mixes over several numbers with no dominant pure play.

If pressed for a heuristic: 2 and 3 carry most weight (they beat the aggressive high numbers on the difference-1 rule).

Takeaway: Acknowledge it's a mixed-strategy equilibrium; walk through why pure play fails.

Q17. Trade on it: I'll pay you the number of H in 10 flips at \$5 per H... wait, I mean I sell you each H at \$5. Deal?

Answer: $E[H] = 5$, so zero EV at \$5 - fair but with no edge and variance, decline as a risk-neutral trader with transaction costs.

If price were \$4.80 you buy; quantify edge = 0.2 per flip-set and relate to your Polymarket threshold of 3 percentage points.

Takeaway: Trick phrasing happens; slow down one second, restate the payoff to confirm, then answer.

5. YOUR STORIES (90 SECONDS EACH, REHEARSED)

A. Polymarket systematic book (HKU team). Found >3pp price-vs-outcome gaps by topic and time-to-resolution; built the scan-rank-limit-execute pipeline; tracked realized P&L. Frame as: edge discovery, execution discipline, P&L accountability - exactly a systematic trading loop.

B. Hang Seng rebalancing prediction (Dymon Asia). Rulebook to point-in-time signals, 82.6% action-level hit rate on 224 events, auditable pipeline, LLM-assisted screening. Frame as: alternative data, backtesting rigor, leakage control - exactly quant research craft.

C. Lightning Network MDP (Oxford). Stochastic control, value iteration - your answer to 'hardest math you've done for fun'.

Behavioral prep: why SIG (entrepreneurial desks, education program, HK placement goal); a time you were wrong (have one with numbers); how you handle losing streaks (risk limits from your Polymarket book).

6. STRATEGY FOR THE DAY

Before: sleep, warm up 20 minutes of mental math the morning of (not hours), re-read this sheet once.

During: think aloud in structured steps (assumptions, method, arithmetic, sanity check). If stuck, restate the problem - half the time the restate reveals the trick.

Speed vs accuracy: they interrupt to add twists; expect the question to change mid-answer - that is a feature, not an attack.

Hong Kong placement: say explicitly: 'My goal is the Hong Kong office - it is where I studied and interned, and I hold an IANG visa with unrestricted work rights.' (JD lists HK placement + sponsorship as standard.)

Ask them: 'How do junior researchers get their first live strategy?' or 'How is the education program structured for the HK cohort?'

After: log every question you were asked into your tracker the same day - SIG recycles patterns across rounds.